Requirements for B-ISDN signalling — broadband services

R J Morley and R R Knight

The broadband ISDN (B-ISDN) is being tailored to become the universal (standardised) future network, and will be capable of supporting a wide range of multimedia, multi-party applications. It is based upon the same principles as its narrowband predecessor and hence can be regarded as a natural extension of narrow ISDN. However, the move away from constant bit rate circuit switching towards the use of asynchronous transfer mode (ATM) technology has provided much needed flexibility, especially in terms of the connection bandwidths and quality of service available to the user.

This paper briefly describes the network capabilities that B-ISDN should support and how they have been derived from a representative sample of user applications principally proposed by ITU-T/ETSI and Digital Audio-Visual Council (DAVIC). The identification of the required network capabilities is the first step towards the specification of signalling protocols for the B-ISDN which must be flexible enough to support the envisaged wide range of current and future advanced applications and services. One such potential future B-ISDN application, which demonstrates the range of signalling functionality required, is the 'Travel Agent Service' which is treated in detail.

1. Introduction

Broadband ISDN (B-ISDN) is being positioned and developed as a single network which is capable of supporting a wide range of audio, video and data applications [1]. However, B-ISDN can only justify its introduction as the preferred network for supporting such a diverse range of applications if it offers advantages in terms of cost, convenience and flexibility, over the reuse of existing networks, to both user and network operator alike. B-ISDN must clearly, therefore, offer more than just a straightforward evolution of the point-to-point 64 kbit/s circuit and packet transfer modes provided by its narrowband predecessor.

The choice of asynchronous transfer mode (ATM) technology as the underlying transport technology for the B-ISDN provides some specific, advantageous facilities over previous network offerings. The most profound of these are a highly flexible network access (due to the cell transport and transfer concepts), fine granularity, and on-demand bandwidth allocation which is independent of the means of transport at the physical layer. However, the presence of ATM transport alone is not enough if B-ISDN is to become the future preferred universal network choice. Intelligence and signalling will need to be added to provide advanced network control, along with powerful operations and maintenance, and network management facilities, if B-ISDN is to fulfil its ambitions.

This paper looks at the initial potential user applications proposed for the B-ISDN and then shows how they have been characterised to derive a set of B-ISDN network services to support them. These services are the starting point for identifying the key network capabilities that B-ISDN signalling systems must support and the ITU-T I.130 method for deriving them is briefly explained. This explanation is followed by a description of the key connection types, identified as part of an overall B-ISDN network capability set, and, finally, section 6 illustrates the use of these connection types with a potential B-ISDN travel agent service.

2. Potential B-ISDN user applications

Advances in data and image processing capabilities have resulted in a plethora of advanced applications, potentially requiring the high bandwidth and flexible network connectivity promised by B-ISDN. The number of such applications is practically endless and is limited only by our imagination and technological constraints, particularly in terms of affordable transmission bandwidth, etc. So far, the majority of applications identified for B-ISDN have been audio-video based but these are not viewed as the only service drivers shaping the definition of B-ISDN network capabilities. Several organisations, such as the International

Telecommunication Union — Telecommunication Sector (ITU-T), the European Telecommunications Standards Institute (ETSI) and the Digital Audio-Visual Council (DAVIC) [2], have been examining what range of applications could conceivably use the B-ISDN and a representative sample is shown in Fig 1.

Although ITU-T defines broadband, in Recommendation I.113 [3], as offering connectivity at greater than 2 Mbit/s, the B-ISDN is by no means limited to only supporting applications requiring connectivity at 2 Mbit/s and above. Building security and traffic monitoring applications can work with connection bandwidths of 64 kbit/s or lower,

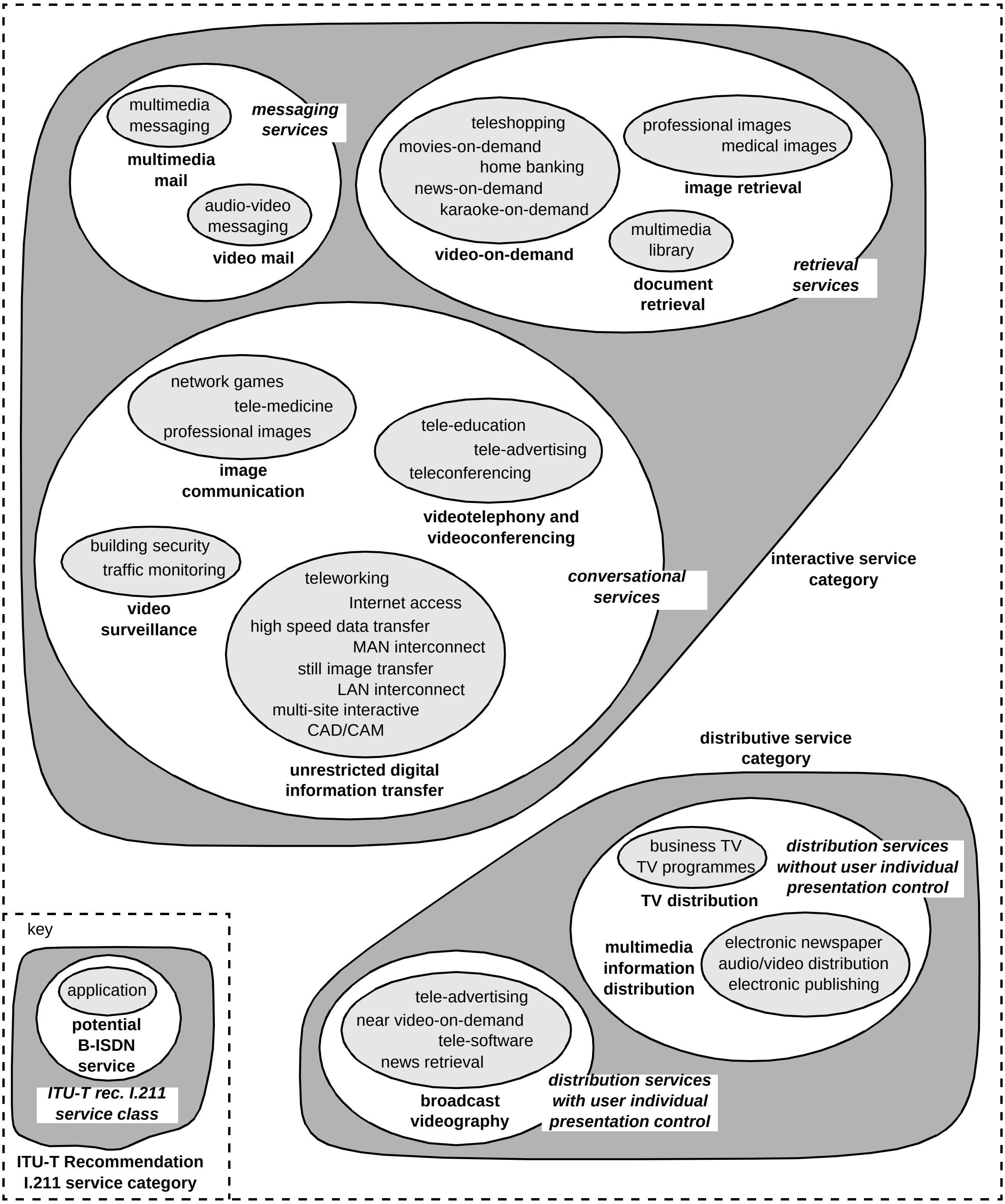

Fig 1 Potential B-ISDN applications, service classes and service categories.

depending upon how frequently information updates are required.

3. Network signalling services

Although the B-ISDN must be capable of supporting a wide range of applications, it cannot offer each application an individual set of connectivity services. Such an approach would result in a separate set of B-ISDN network capabilities for each application and a separate signalling protocol for those applications which wanted ‘on-demand’ connectivity. In order, therefore, to keep the range of network capabilities that are needed down to a manageable set, a smaller sub-set of B-ISDN services must be defined, from which the network capabilities can be derived, and which can support a group of applications sharing some common characteristics. ITU-T Recommendation I.211 [4] classifies services into service categories and service classes. The service categories described within Recommendation I.211 have also been included in Fig 1 together with the service classes, within each category, relevant to the applications mentioned. The specification of the I.211 service classes, and the information types they contain, e.g. audio or video, was the first step taken by ITU-T towards defining the limited set of B-ISDN services that should be supported. If, for example, as shown in Fig 1, B-ISDN could support a video-on-demand service, then it can support tele-shopping, karaoke-on-demand and other multimedia interactive applications. Although I.211 contains a comprehensive set of B-ISDN services, it is by no means complete and must evolve to meet the needs of emerging applications.

3.1 Bearer and teleservices

From a signalling perspective, the limited set of services to be supported by the B-ISDN may be internationally standardised and offered by an administration as bearer services or teleservices, as illustrated in Fig 2. A bearer service is assumed just to provide connectivity and takes no account of, nor offers support for, an application. The ATM adaptation layer (AAL) could be empty with the cell payloads, for example, being passed directly to the application. However, it is more common to include the

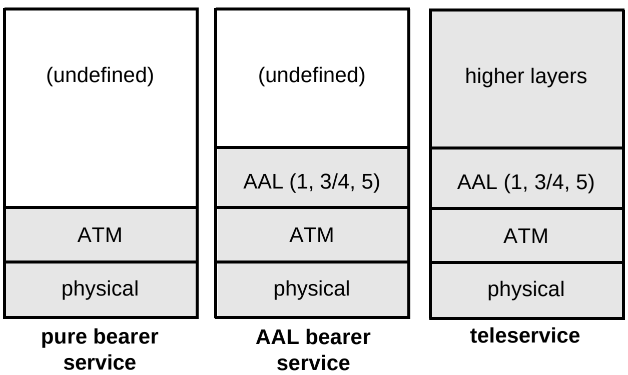

Fig 2 Bearer and teleservices in the B-ISDN.

specifics of the AAL within the bearer service, especially if the B-ISDN is required to support interworking between networks. A telecommunications service, or teleservice, on the other hand, requires all application characteristics, expressed in the International Standards Organisation’s (ISO’s) open systems interconnection (OSI) seven-layer protocol model [5], to be specified. The list of standardised bearer and teleservices identified so far for the B-ISDN, or which could be supported by the B-ISDN, are shown below:

- bearer services currently defined for the B-ISDN:
  - — broadband connection-oriented bearer service,
  - — broadband connectionless data service,
  - — virtual path service for reserved and permanent communications,
- B-ISDN standardised teleservices:
  - — teleconferencing,
  - — audiographic conference,
  - — videotelephony,
  - — videoconferencing,
  - — broadband videotex,
  - — multimedia conferencing,
  - — audiovisual interactive services,
  - — telewriting,
  - — broadband teletex,
  - — high-speed telefax.

The involvement of the network in a teleservice depends upon the degree to which it has been standardised. Four levels of teleservice standardisation have been proposed [6] for the B-ISDN and they are described below.

Standardised teleservice

All of the attributes of the standardised telecommunications service assume standardised, default values which cannot be changed. The network is assumed to allocate all of the resources for the service and ensures that the semantic rules given to the service are obeyed. In addition, the network will also ensure terminal-to-terminal compatibility relating to higher layer functions (HLFs) and lower layer functions (LLFs) of the teleservice. The lower layer functions cover the physical layer, ATM layer and AAL aspects of the service. The higher layer functions cover all aspects of the service above the AAL functionality. If the standardised service is a bearer service, the network will only ensure LLF compatibility. ITU-T has undertaken standardisation activity on several multimedia target teleservices for the B-ISDN as itemised above.

Although strictly confined to the functionality defined within the standards-based service description, standardised teleservices can prevent the proliferation of a large number of standards for multimedia information types. However, if a change is required to a standardised service, this can only be done via the appropriate standards body, which can take a long time relative to the speed at which applications are being developed.

Semi-standardised teleservice

Semi-standardised teleservices permit a greater degree of flexibility over fully standardised services. Semi-standardised services allow the user to select or discard some parts of the service according to a general, standardised service description. Within a multimedia context, two types of flexibility are possible — selection of a specific value of an attribute from a set of allowed values, and/or selection of which parts of a multimedia service should be used in an actual call. An example of the first kind of flexibility could be the selection of the bandwidth value for a data transfer operation. The second kind of flexibility could allow a user to select the audio but not the video component of a videotelephony service.

The user is able to select the required service configuration at call set-up and/or during the lifetime of the call, and the network, in response, allocates all of the necessary resources. The network will also check for terminal-to-terminal compatibility relating to both the higher and lower layer of a teleservice or just the lower layer functions of a bearer service.

The presence of semi-standardised services starts to give the user some choice over what aspects of a teleservice or bearer service are used. However, they are still reliant on standards bodies to introduce new service options or to modify existing ones.

User-defined teleservice

This teleservice category allows users to predefine the characteristics of a service which is required *a priori* with the network provider. Once predefined, users can then establish actual calls using their own user-defined service. The network will allocate all of the required network resources for the call, but is not responsible, in principle, for ensuring complete terminal-to-terminal compatibility. However, the network may be responsible for ensuring terminal compatibility relating to individual standardised components of the service. Although user-defined teleservices can only be employed by customers who have previously agreed the service definition with the network, the service required for an application can be built from a set of standardised and user-defined components. The network operator only has control over parts of the service based upon standardised components, with respect to compatibility checking, etc.

Open teleservice

The open teleservice permits a user to have a service consisting of standardised and non-standardised components. The selection of the required components is specified only at call set-up time and can be modified during the lifetime of a call.

The network will allocate all of the required resources for the call but the user is responsible for terminal-to-terminal compatibility. The network may, however, assist in the compatibility checking related to standardised components used for the call.

The open service allows the user to obtain the desired service from the network as and when required. The user is also able to freely modify the characteristics of an active call as desired, and only makes use of the network resources and service elements that are really required.

3.2 B-ISDN supplementary services

An additional ‘supplementary’ set of B-ISDN service functionalities has also been defined and standardised by ITU-T to complement the specified range of bearer services and teleservices. Supplementary services cannot exist on their own and hence are used in conjunction with a bearer service or teleservice, either on a per-call basis or under a subscription option. The existence of supplementary services offer to the application additional information or functionality which can be used to create value-added features, such as notification to the user of who is calling. The following list shows the B-ISDN supplementary services currently defined by ITU-T (this supplementary service set was based upon those previously defined for the narrowband ISDN):

- calling line identification presentation,
- calling line identification restriction,
- connected line identification presentation,
- connected line identification restriction,
- closed user group,
- priority call,
- sub-addressing,
- direct dialling in,
- multiple subscriber numbering,
- user-to-user signalling.

4. ITU-T Recommendation I.130 method for signalling protocol specification

The textual definition of the bearer and teleservices to be supported by a B-ISDN are themselves insufficient to identify the network capabilities required and hence define suitable signalling protocols. A rigorous approach is required, building upon the textual descriptions, which allows each service to be examined in a common way and its requirements expressed within a common set of functions, service components and attributes. The job of specifying such an approach was undertaken by ITU-T and their resultant three-stage method has been embodied within Recommendations I.130 [7], I.140 [8] and I.210 [9]. The I.130 method, although originally aimed at narrowband ISDN services, has been shown [6] to be extendible for capturing service characteristics and behaviour of many multimedia services identified for the B-ISDN [10]. The first of these three stages involves service definition from the user's viewpoint and it has been resolved into three steps.

- Prose description

  This extends the textual description using a template to capture the involvement of the service users with each other, and the types of user plane information that flows between them.

- Static description

  This identifies as far as possible a complete set of common connection and service user attributes. The characteristics of the service captured within the prose description are then used to derive values for the common set of connection and service attributes to create a static instance of the service to be described. This step specifies the service entities, attributes and attribute values to be carried by the signalling system.

- Dynamic description

  This captures the control plane information that must flow between users and the network to create, modify and delete an instance of the service.

Stages two and three of the I.130 are then used to help formulate how the network can support the static and dynamic behaviour defined in stage one. Stage two derives the network functional model required and the information flows that must exist between a number of functional entities defined within the functional model. Stage three then maps the functional entities defined in stage two on to physical control plane network entities (ATM switches) and deals with the derivation of the signalling protocols, based upon the information flows that must take place over any external interfaces between physical network entities. In common with narrowband signalling systems, the physical interfaces requiring a signalling protocol identified so far within B-ISDN are between the user and the network, and between network nodes (i.e. ATM switches).

5. Identification of network capabilities

An analysis of the potential B-ISDN bearer and teleservices within ITU-T [9, 11] has so far identified five connection types, on a switched, semi-permanent and permanent basis, that the B-ISDN should support from a signalling point of view. These connection types offer bi-directional point-to-point, unidirectional point-to-multi-point, unidirectional multipoint-to-point, bi-directional point-to-multipoint and multipoint-to-multipoint connect-ivity. It was realised, however, that further network capabilities will be incorporated into B-ISDN, in evolutionary steps, to meet new user requirements and accommodate advances in network developments and progress in technology [12]. Ideally, connections must support both circuit-mode and packet-mode services of a monomedia and/or multimedia type and of a connectionless or connection-oriented nature and in a bi-directional or unidirectional configuration.

While all five types are identified in this paper, only point-to-point and unidirectional point-to-multipoint con-nections (type 1 and type 2) are currently supported in the signalling procedures for B-ISDN signalling CS-2. However, no paper on the subject of ATM signalling requirements would be complete without a description of the complete signalling requirements for connection control laid down by ITU-T in the B-ISDN CS-2 requirements. A brief description of each of the five connection types is therefore given in the following sub-sections.

5.1 Point-to-point connections

The simplest category is a point-to-point connection, which is a single switched link, i.e. an ATM virtual channel connection, between two end-points. This is referred to as a type 1 connection and may be either bi-directional or unidirectional. Bi-directional connections are permitted to have an asymmetrical bandwidth, but must have the same ATM layer transfer capability and AAL type in both directions.

5.2 Unidirectional point-to-multipoint connections

Unidirectional point-to-multipoint connections, type 2 (see Fig 3), utilise the ATM switching fabric's capability to replicate ATM cells and to direct the resulting streams to different end-points. This function is analogous to the broadcast of information, since one source may transmit information to many receivers. However, within ATM studies, this type of capability is more usually referred to as a multicast connection since the number of receivers is limited by the number of connections possible within the network. An information broadcast capability implies an unlimited number of receivers and hence unlimited network connection resources. Type 2 connections are restricted to a unidirectional capability, having only a single source and multiple sinks.

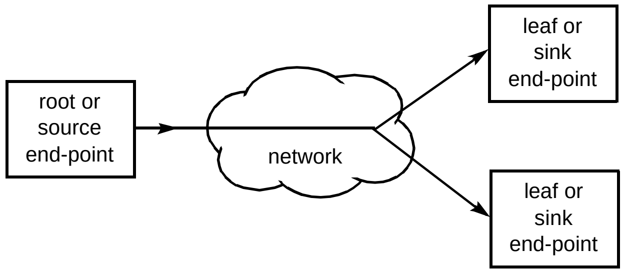

Fig 3 Unidirectional point-to-multipoint connection (type 2).

5.3 *Unidirectional multipoint-to-point connection*

Unidirectional multipoint-to-point connections, type 3, provide unidirectional multipoint-to-point virtual channel connections. They are currently only supported in AAL type 3/4 [13] and require an underlying transport system that is capable of multiplexing two or more connection sources into a single connection for reception by a single sink, as shown in Fig 4.

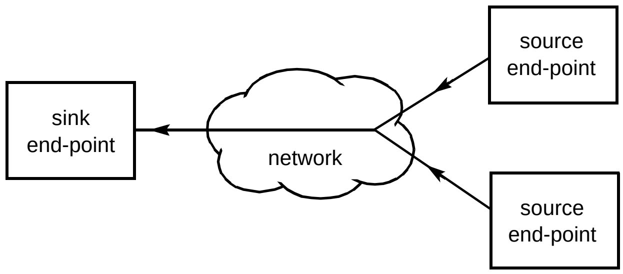

Fig 4 Unidirectional multipoint-to-point connection (type 3).

5.4 *Multipoint-to-multipoint connections*

Multipoint-to-multipoint connections, type 4, perform multipoint to multipoint functionality — a complete communications connection supporting more than two end-points, essentially in a star configuration. Any information sent by an end-point is received by all the other end-points, although each end-point only has a single bi-directional connection. There are some restrictions on the implementation of this type of connection in the AAL. AAL type 3/4 can support this type of connection, although it may also be required in other AALs for some services. It is recognised that to meet this requirement, equipment may need to be provided within the network to perform the multiplexing, rather than using the capabilities of the AAL. This equipment may be referred to as a conference bridge — but this is a rather abused term, since this is generally understood to be a piece of equipment that is providing some conferencing service, rather than merely multiplexing and de-multiplexing ATM cells. The functional view of the connection is shown in Fig 5, though this does not address the issue of how the connection should be implemented.

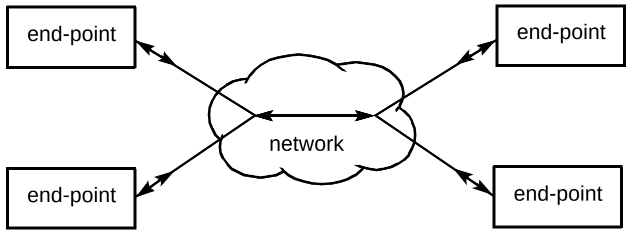

Fig 5 Multipoint-to-multipoint connection (type 4).

5.5 *Bi-directional point-to-multipoint connections.*

Type 2 and type 3 connections (point-to-multipoint and multipoint-to-point) are unidirectional. Bi-directional point-to-multipoint connections, type 5, are defined as type 2 connections overlaid with a type 3 to provide a bi-directional connection. The introduction of a bi-directional capability allows leaves in a distribution tree to have a ‘back channel’ which may be used to send information to the root. Type 5 connections can support such applications as broadcast TV with a voting mechanism since half of the connection from the leaf to the root can be used to multiplex, on an AAL type 3/4 level, the users’ votes going back to the root. The connection topology is shown in Fig 6, and is always referred to as a bi-directional point-to-multipoint connection, although the terminology bi-directional multipoint-to-point connection should be equally valid.

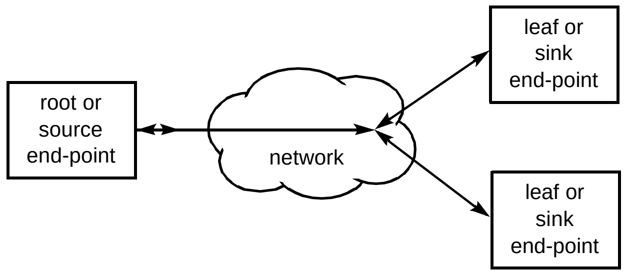

Fig 6 Bi-directional point-to-multipoint (type 5).

5.6 *Mixture of connection types*

The CS-2 requirements would be regarded as being fully met when a complex call may be established consisting of any mixture of type 1 and type 2 connections, as represented by the configuration shown in Fig 7. This should include a party which may choose to participate in the call, but which is not attached to any connections — and therefore does not directly communicate with any of the other parties (i.e. has no user plane communications). All parties and connections within the call may be replicated (i.e. there may be more than one connection of type 2 (labelled ‘a’ in Fig 7), in which case there may be more parties who are type 2 source end-points).

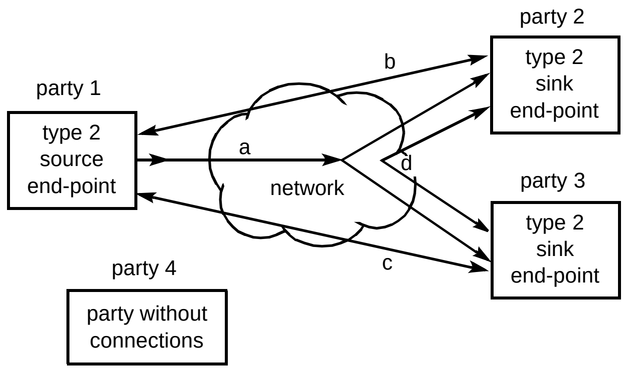

Fig 7 A complex CS-2 call from the signalling requirements.

5.7 Third party set-up

The B-ISDN CS-2 requirements also identify the need to support an end-point’s ability to set up a connection between two other end-points, without the connection owner being involved in the sending or receiving of any information over that connection. This type of facility is known as a third party set-up, and is shown in Fig 8.

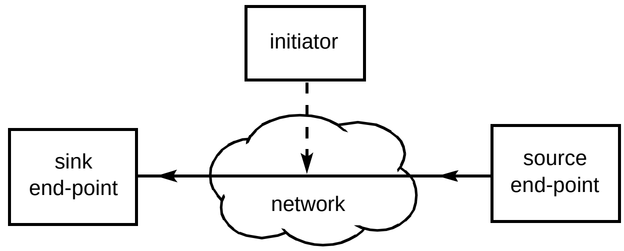

Fig 8 Third party set-up.

6. The Travel Agent Service

This service scenario has been included as a theoretical example of an open teleservice which B-ISDN’s future capability needs to be able to support. It illustrates a number of problems for B-ISDN signalling since it must support and integrate, on demand, several different teleservices. Taking part in the service are four end users — two service providers, a retailer and a customer. It is assumed that all have some form of an intelligent broadband terminal.

6.1 Scenario

The service starts with the customer contacting the travel agent with a view to booking a holiday. The initial communication is telephony (speech only), but, since both parties have broadband terminals, this initial communication is upgraded to videotelephony. The travel agent determines the customer’s requirements, such as price restrictions, type of holiday, general region, and then initiates a search of available holidays with the travel company’s ‘agents only’ database. The results of the search are then provided to the customer as data, and the customer selects those that look promising. The travel agent then offers to provide video clips to the customer of the selected holidays — both resorts and hotels could conceivably be available. However, the customer’s network connection could be xDSL based and consequently may not have enough bandwidth to simultaneously support a video connection, for viewing the holiday destinations, and a videotelephony connection, for communicating with the travel agent. Therefore, the videotelephony link to the travel agent must be dropped but the travel agent still needs to remain in the call, however, to be reconnected at the end of the video clips.

The customer, in this theoretical scenario, is wholly convinced that one of the holidays will be suitable and decides to book. The travel agent must make the booking with the travel company — an automatic transaction on a data link — and then ask the customer for payment. Fortunately, the theoretical customer has an acceptable credit card, but, since this is a late booking, must pay the entire amount. A double link to the credit card company is therefore required, enabling the travel agent to provide the retail outlet number and receive an authorisation code. The customer meanwhile must provide the important PIN or similar form of electronic signature to confirm the payment.

6.2 Communication configurations

From the network point of view, the call begins with a single telephony link between two parties and is then upgraded to a videotelephony link. The signalling requirements are therefore to provide a single type 1 connection, between two parties, and subsequently allow an on-demand modification of the connection’s traffic characteristics in order to support the extra bandwidth requirements of videotelephony (Fig 9).

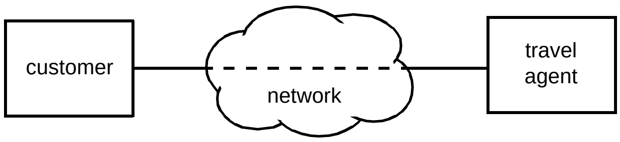

Fig 9 Initial configuration.

The introduction of data links to the travel company enables the travel company’s database to be searched and the results provided to the customer. This requires the addition of a third party and two connections (both data types). The signalling requirements must consequently allow upgrade of the connection from a two-party call to a multiparty call in which each party is connected to each other using a type 1 connection. The call must allow each connection to have different ATM layer transfer capabilities (such as continuous bit rate/variable bit rate) and bandwidths (Fig 10).

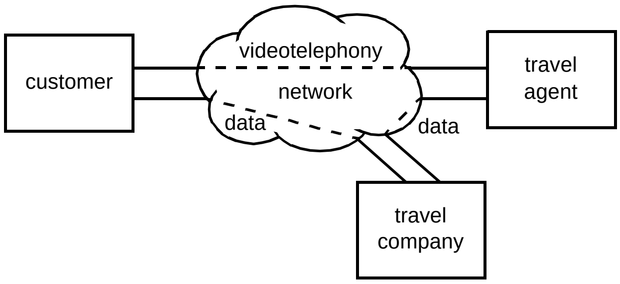

Fig 10 Configuration for searching the travel company's database.

Having found some suitable choices, the travel agent offers the customer video clips to enable a final selection. Assuming that the customer's access point has limited bandwidth, the size of the connection between the customer and the travel agent has to be reduced, with a consequential downgrading of the teleservice from videotelephony back to telephony.

At the same time, the video server is introduced into the call, and the customer's available bandwidth is reassigned from the travel agent's connection to the new connection with the video server (Fig 11).

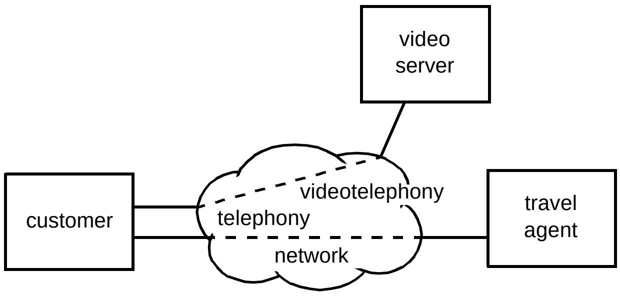

Fig 11 Configuration for viewing the video server clips of the holiday destination.

The signalling requirements of this portion of the call are quite complex. As well as the multiparty aspects of adding a new customer, the transfer of bandwidth from one connection to another is a radical new requirement. If the bandwidth is not retained by the customer, it may be seized by another application in the same household.

Once the final holiday has been selected and the video clips have finished, the call needs to revert back to the original configuration shown in Fig 9. The holiday is then booked with the travel firm and payment is expedited by another data connection to the credit card company, enabling the travel agent to enter the details and the customer confirm the transaction using a PIN or password (Fig 12).

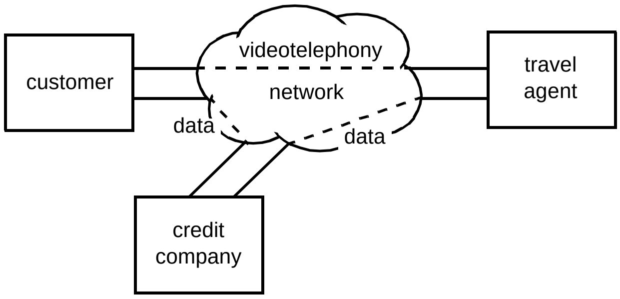

Fig 12 Configuration for making payment.

The signalling requirements for making payment are the same as those described above for searching the travel agent's database of holiday destinations (Fig 10).

7. Conclusions

Consideration of the types of application that B-ISDN is to support is an essential first step towards the definition of the bearer, teleservice, and supplementary services that the network needs to support. From these bearer services, teleservices and supplementary services, the network capabilities needed, which must ultimately be supported by the signalling systems, can be derived. So far, the B-ISDN signalling systems that have been produced can support bearer services, teleservices and supplementary services needing point-to-point or unidirectional point-to-multipoint network connections. Clearly, these capabilities are insufficient to accommodate the applications cited above and hence further work towards the target solution for B-ISDN signalling is taking place, based upon the B-ISDN signalling CS-2 work [11].

There is not perceived to be a single killer application or service that B-ISDN should be tailored to support. B-ISDN can only be successful if it is developed as a general-purpose platform, capable of supporting a variety of applications, such as the travel agent service. Of key importance for B-ISDN will be the ability to seamlessly interwork with or support, in a complementary manner, similar/identical applications that can run over the Internet.

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13 ITU-T Recommendation I.363.3: 'B-ISDN ATM adaptation layer specification: type 3/4 AAL', (August 1996).

Roger Morley joined BT in 1980 as an apprentice in Nottingham. Upon completing his apprenticeship in 1983, he undertook a variety of technician jobs. In 1985 he was sponsored by BT to read for a BEng degree in Telecommunications Engineering at Queen Mary College, London. Upon graduation in 1988, he then worked at BT Laboratories on a PC-based X.21 communications card for use with BT's pilot ISDN service. In 1991 he then moved into BT's ISDN Services and Applications Laboratory where he developed notable expertise in ISDN signalling systems, Group 3/4 facsimile and the end-to-end testing of international ISDN links. In 1993, he gained an MSc degree in Telecommunication and Information Systems from the University of Essex. In 1994, he moved over to work on B-ISDN service definition within RACE MAGIC, which was specifying signalling systems for advanced, multimedia, multi-party services for release 2/3 of B-ISDN. Following the completion of RACE MAGIC in 1994, he was assigned to internal projects concerned with introducing signalling and intelligence into BT's future broadband networks. He is also an Associate Member of the IEE.

Dick Knight joined BT as an apprentice in 1971 and moved to BT Laboratories in 1979. He worked on System X development in the areas of the test network subsystem and customer line card design acceptance testing. In 1986 he was made responsible for the software to implement a programmable ISDN signalling design acceptance tester. He designed and implemented a protocol to collect network management information for the Jetphone network. From 1993 to 1994 he moved onto broadband signalling design by managing the collaborative RACE MAGIC project on behalf of twelve European companies — a project whose results are still having one of the largest impacts on broadband signalling design in the world. With a short break in 1996 to design and patent an application protocol to collect information such as burglar alarms over the ISDN D-channel packet access, he has continued to work on various aspects of broadband signalling, both internal projects and representing BT at ETSI SPS5 with the occasional representation at ITU-T to contribute some of the MAGIC results.